

Probability distributions for continuous random variable

Jean Paul Nsabimana

March 30, 2026

Continuous Probability Distributions

1. Continuous Uniform Distribution

The density function of the continuous uniform random X on the interval $[a, b]$:

$$f(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise.} \end{cases}$$

The cumulative density function $F(X)$ is

$$F(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & x > b. \end{cases}$$

The moment generating function of the uniform distribution is given by

$$G_x(t) = E(e^{tX}) = \frac{e^{tb} - e^{ta}}{t(b-a)}.$$



Fig4.2.Graphic of $f(x)$

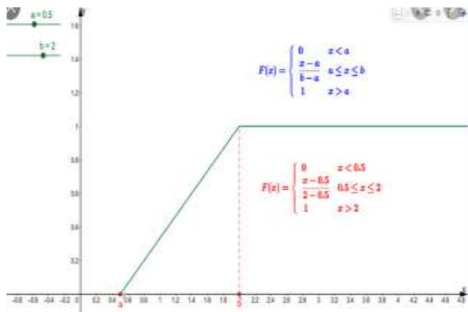


Fig4.3: Graphic of $F(x)$

Proposition

The mean and variances of the uniform distribution

$$E(x) = \frac{a+b}{2} \quad \sigma^2 = \frac{(b-a)^2}{12}.$$

Example

Suppose large conference room for a certain company can be reserved than 4 hours. However, the use of the conference room is such that both long and short conferences occur quite of often. In fact, it can be assumed that length X of a conference has a uniform distribution on the interval $[0, 4]$.

- 1 What is the probability density function?
- 2 What is the probability that any given conference lasts at least 3hours.
- 3 What are the mean and variance.

Answer

- ① The appropriate density function for the uniformly distributed random variable X in this situation is

$$f(x) = \begin{cases} \frac{1}{4} & 0 \leq x \leq 4 \\ 0 & \text{otherwise.} \end{cases}$$

- ② The probability that any given conference lasts at least 3 hours is given as

$$P(X \geq 3) = \int_3^4 \frac{1}{4} dx = \frac{1}{4}.$$

- ③ The mean and variance values are calculated as

$$\begin{aligned} E(x) &= \frac{a+b}{2} = \frac{0+4}{2} = 2 \\ \sigma^2 &= \frac{(b-a)^2}{12} = \frac{(4-0)^2}{12} = \frac{16}{12} = \frac{4}{3}. \end{aligned}$$

2. Exponential distribution.

The continuous random X has an exponential distribution with parameter λ , if its density function is given by

$$f(x, \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0. \end{cases}$$

The cumulative density function $F(X)$ is given by:

$$F(x) = \begin{cases} 1 - e^{-\lambda x} & x \geq 0 \\ 0 & x < 0. \end{cases}$$

The moment generating function of the exponential distribution is given by

$$G_x(t) = E(e^{tX}) = \frac{\lambda}{\lambda - t}.$$

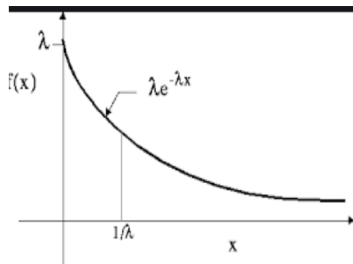


Fig4.4: pdf for exponential distribution

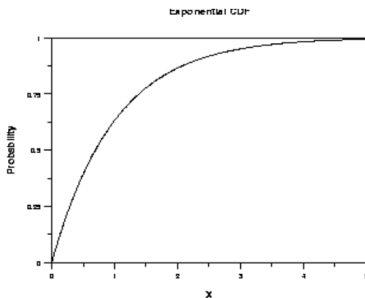


Fig4.5 :CF for exponential distribution

Proposition

The mean and variance of the exponential distribution are

$$E(X) = \frac{1}{\lambda}; \quad \sigma^2 = \frac{1}{\lambda^2}.$$

Example

Suppose that a system contains a certain type of component. Whose time n year, to failure is given by X . The random variable X is modeled nicely by the exponential distribution with mean time to failure exponential distribution is $\lambda = \frac{1}{5}$. If 5 of these components are installed in different systems, what the probability that at least 2 are still functioning at the end of 8 years.

Answer:

The probability density function is given by:

$$f(x, \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0. \end{cases} = \begin{cases} \frac{1}{5} e^{-\frac{1}{5}x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

The probability that a given component is still functioning after 8 years is given

$$P(X > 8) = \frac{1}{5} \int_8^{+\infty} e^{-\frac{1}{5}x} dx = e^{-\frac{8}{5}} \approx 0.2.$$

3. The Normal Distribution

Def

It is a special distribution that we will use just about every day for a continuous random variable, has the following properties:

Properties

- 1 It can take on any value (not just integers, as do the binomial and Poisson distribution)
- 2 It is symmetric about the mean, μ
- 3 The standard deviation of the distribution is symbolized by σ , which is the horizontal distance between the mean and the point of inflection on the curve.
- 4 It approaches the horizontal axis on both the left and right side without touching, that is the x-axis is a asymptote.
- 5 It is bell shaped with transition points one standard deviation from the mean.

Properties cont ...

- 1 Approximately 68% of the data points lie within one standard deviation of the mean.
- 2 Approximately 95% of the data points lie within two standard deviations of the mean.
- 3 Approximately 99.7% of the data points lie within three standard deviations of the mean.
- 4 The total area under the curve and above the horizontal axis is equal to 1.
- 5 Since it is symmetrical distribution, half the area is on the left of the mean and half is on the right

Figure

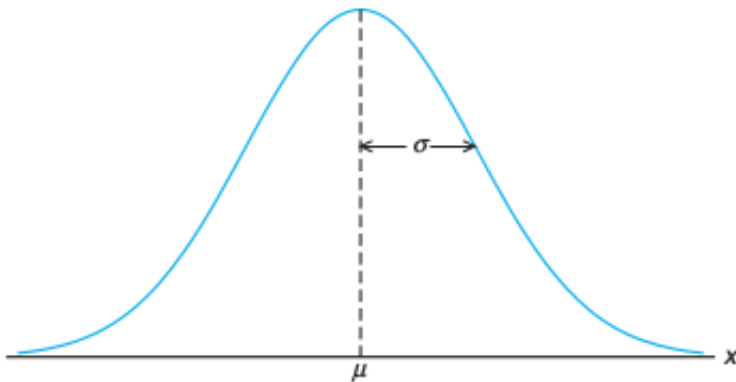


Figure: Normal curve

Formula

Given a normal random variable X , with mean μ and variance σ^2 that can take on any value between negative and positive infinity ($-\infty$ to $+\infty$), normal distribution formula is as follows:

$$P(x; \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(x-\mu)^2}, -\infty < x < \infty$$

where $\pi = 3.14159\dots$ and $e = 2.71828\dots$

The Standard Normal Distribution

Def

The Standard Normal distribution follows a normal distribution and has mean 0 and standard deviation 1.

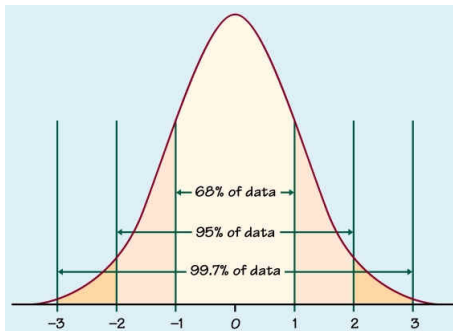


Figure: Standard normal distribution

Note

Notice the Figure 2 is perfectly symmetric about 0. If a distribution is normal but not standard, we can convert a value to the Standard normal distribution Z score by finding first as how many standard deviations away the number is from the mean. The number of standard deviations from the mean is called the *z-score* and can be found by the formula:

$$z = \frac{x - \mu}{\sigma}$$

The Z Score and Area

Often we want to find the probability that a z-score will be less than a given value, greater than a given value, or in between two values. To accomplish this, we use the Table from the textbook and a few properties about the normal distribution.

Example 1

Given a standard normal distribution, find the area under the curve that lies

- 1 to the right of $z = 1.84$ and
- 2 between $z = -1.97$ and $z = 0.86$.

Solution

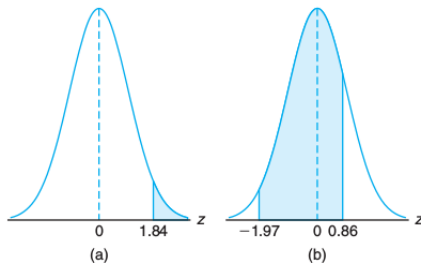


Figure: Area under the curves for example 1.

Solution cont ...

- (a) The area in Figure 3(a) to the right of $z = 1.84$ is equal to 1 minus the area in Table A.3 to the left of $z = 1.84$, namely, $1 - 0.9671 = 0.0329$.
- (b) The area in Figure 3(b) between $z = -1.97$ and $z = 0.86$ is equal to the area to the left of $z = 0.86$ minus the area to the left of $z = -1.97$. From Table A.3 we find the desired area to be $0.8051 - 0.0244 = 0.7807$.

Example 2

Given a standard normal distribution, find the value of k such that

① $P(Z > k) = 0.3015$ and

② $P(k < Z < -0.18) = 0.4197$.

Example 2

Given a standard normal distribution, find the value of k such that

- ① $P(Z > k) = 0.3015$ and
- ② $P(k < Z < -0.18) = 0.4197$.

solution

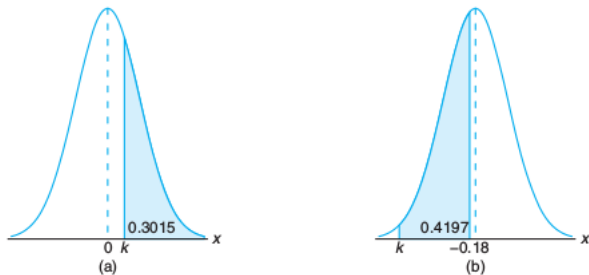


Figure: Area under the curves for example 2.

Example 2 cont...

Solution cont ...

- (a) In Figure 4(a), we see that the k value leaving an area of 0.3015 to the right must then leave an area of 0.6985 to the left. From Table A.3 it follows that $k = 0.52$.
- (b) From Table A.3 we note that the total area to the left of -0.18 is equal to 0.4286. In Figure 4(b), we see that the area between k and -0.18 is 0.4197, so the area to the left of k must be $0.4286 - 0.4197 = 0.0089$. Hence, from Table A.3, we have $k = -2.37$.

Example 3

Given a random variable X having a normal distribution with $\mu = 50$ and $\sigma = 10$, find the probability that X assumes a value between 45 and 62.

Example 3

Given a random variable X having a normal distribution with $\mu = 50$ and $\sigma = 10$, find the probability that X assumes a value between 45 and 62.

solution

The z values corresponding to $x_1 = 45$ and $x_2 = 62$ are

$$z_1 = \frac{45 - 50}{10} = -0.5 \quad \text{and} \quad z_2 = \frac{62 - 50}{10} = 1.2 \quad \text{respectively}$$

Therefore, $P(45 < X < 62) = P(-0.5 < Z < 1.2)$.

Figure

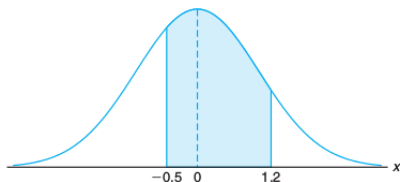


Figure: Area under the curve for example 3.

Cont...

$P(-0.5 < Z < 1.2)$ is shown by the area of the shaded region in Figure 5. This area may be found by subtracting the area to the left of the ordinate $z = -0.5$ from the entire area to the left of $z = 1.2$. Using Table A.3, we have $P(45 < X < 62) = P(-0.5 < Z < 1.2) = P(Z < 1.2) - P(Z < -0.5) = 0.8849 - 0.3085 = 0.5764$.

Exercise 1

The average salary for first-year teachers is \$27,989. Assume the distribution is approximately normal with standard deviation \$3250.

- 1 What is the probability that a randomly selected first-year teacher makes between \$20,000 and \$30,000 each year?
- 2 What is the probability that a randomly selected first-year teacher has a salary less than \$20,000?

Exercise 2

Test scores for a given class are normally distributed with a mean of 80 and standard deviation of 12, with several over 100 (with bonus points). Find the probability that a randomly selected student from this class has a test score that is...

- ① less than 92
- ② between 80 and 90
- ③ greater than 70

Exercise 3

Find the number b that satisfies:

① $P(z < b) = 0.9664$

② $P(z < b) = 0.3050$

③ $P(z > b) = 0.0078$